

AI Market Intelligence Terminal

Data Sources Map — what we collect and where from

Legend: [FREE] Free API [PAID] Paid / free tier available [COMPLEX] Complex integration

1. Macroeconomic Data

Indicator	What it shows	Source	API / Endpoint	Frequency	Status
US 10Y Yield	Cost of money, risk benchmark	FRED	FRED series: DGS10	Daily	FREE
Real Yield (TIPS)	Inflation-adjusted yield	FRED	FRED series: DFII10	Daily	FREE
2Y Yield	Fed rate expectations	FRED	FRED series: DGS2	Daily	FREE
Yield Curve 2s10s	Inversion = recession signal	FRED	FRED series: T10Y2Y	Daily	FREE
Fed Funds Rate	Current Fed rate	FRED	FRED series: FEDFUNDS	Monthly	FREE
CPI YoY	Consumer inflation	FRED / BLS	FRED series: CPIAUCSL	Monthly	FREE
PCE	Fed preferred inflation gauge	FRED	FRED series: PCEPI	Monthly	FREE
GDP Growth	Economic output growth	FRED	FRED series: A191RL1Q225SBEA	Quarterly	FREE
Unemployment	Labor market health	FRED	FRED series: UNRATE	Monthly	FREE
DXY Index	US dollar strength	yfinance	Ticker: DX-Y.NYB	Real-time	FREE
Gold XAU/USD	Safe haven signal	yfinance	Ticker: GC=F	Real-time	FREE
WTI Oil	Commodity / inflation signal	yfinance	Ticker: CL=F	Real-time	FREE
Buffett Indicator	Market Cap / GDP ratio	FRED + calc	FRED: WILL5000 / GDP	Quarterly	FREE

2. Liquidity Indicators

Indicator	What it shows	Source	API / Endpoint	Frequency	Status
TGA Balance	Treasury cash — spending injects liquidity	Treasury	fiscaldata.treasury.gov/api/v1/accounting/dts/dts_table_1	Daily	FREE

Indicator	What it shows	Source	API / Endpoint	Frequency	Status
Reverse Repo RRP	Excess liquidity parked at Fed — drop = risk-on	NY Fed	markets.newyorkfed.org/read/download_csv	Daily	FREE
Fed Balance Sheet	Size of Fed balance (QT/QE)	FRED	FRED series: WALCL	Weekly	FREE
M2 Money Supply	Total money supply in system	FRED	FRED series: M2SL	Monthly	FREE
MOVE Index	Bond volatility — stress indicator like VIX	ICE / calc	Via options chain or Investing.com scrape	Daily	COMPLEX
Bank Reserves	Commercial bank reserves at Fed	FRED	FRED series: WRESBAL	Weekly	FREE
Credit Spreads HY	HY vs IG — risk appetite signal	FRED	FRED series: BAMLH0A0HYM2	Daily	FREE

3. Market Sentiment

Indicator	What it shows	Source	API / Endpoint	Status
Fear & Greed Index	Composite sentiment score 0-100	CNN	production.dataviz.cnn.io/index/fearandgreed/graphdata	FREE
VIX	S&P; 500 implied volatility	yfinance	Ticker: ^VIX	FREE
AAll Sentiment	Retail investor bull/bear survey	AAll	aall.com/sentiment-survey (weekly scrape)	FREE
Put/Call Ratio	Options market positioning	CBOE	cboe.com/us/options/market_statistics/	FREE
News Sentiment	Headlines: bullish / bearish NLP	RSS parsers	Reuters, FT, Bloomberg RSS feeds	FREE
Reddit WSB	Retail forum sentiment	Reddit API	reddit.com/r/wallstreetbets.json	FREE
Twitter / X	Macro community sentiment	X API v2	api.twitter.com/2/tweets/search/recent	PAID

4. Prediction Markets (real-money forecasts)

Platform	What to track	API / Endpoint	Status
Polymarket	Fed rate decision, recession, geopolitical events	gamma-api.polymarket.com	FREE
Kalshi	Fed, CPI, GDP, unemployment outcomes	trading-api.kalshi.com/trade-api/v2	FREE
PredictIt	Political and macro events	predictit.org/api/marketdata/all	FREE

Platform	What to track	API / Endpoint	Status
CME FedWatch	Fed decision probabilities from futures pricing	cmegroup.com (scrape or calc from futures)	FREE

* Polymarket and Kalshi use real money — probabilities are most accurate

5. Insiders & Institutional Flows

Data	What it shows	Source	API / Endpoint	Frequency	Status
SEC Form 4	Insider buying / selling transactions	SEC EDGAR	efts.sec.gov/LATEST/search-index?q=form-type:4	Real-time	FREE
13F Filings	Hedge fund & big money positions	SEC EDGAR	efts.sec.gov/LATEST/search-index?q=form-type:13F	Quarterly	FREE
13D / 13G	5%+ stakes — activist investor positions	SEC EDGAR	efts.sec.gov — form-type: SC 13D	Event-based	FREE
FINRA Short Data	Large off-exchange dark pool prints	FINRA	regsho.finra.org (daily short volume)	Daily	FREE
Options Flow	Large unusual options bets	Barchart	barchart.com/options/unusual-activity (scrape)	Real-time	COMPLEX

6. Market Cycles

Cycle	Key Indicators	Data Sources	Phases
Business cycle	ISM PMI, unemployment, GDP, earnings growth	FRED, ISM	Expansion > Peak > Contraction > Trough
Credit cycle	HY spreads, lending standards, credit growth	FRED: BAMLH0A0HYM2	Easy credit > Tightening > Stress > Reset
Liquidity cycle	RRP, TGA, Fed balance sheet, M2	NY Fed, FRED	QE > Neutral > QT > Crisis liquidity
Debt supercycle	Debt/GDP ratio, long-term rates, demographics	FRED, BIS	Accumulation > Peak > Deleveraging (Dalio)
Presidential	4-year market pattern relative to US election cycle	Historical data	Yr1 weak > Yr2 bottom > Yr3-4 rally

* Cycle Detector agent determines current phase using combined signals from sections 1-3

Tech Stack Summary

Layer	Technology	Purpose
Backend	FastAPI (Python)	API server, scheduler, agent orchestration
Database	Postgres + pgvector	Timeseries storage + vector embeddings for RAG

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Cache	Redis / Upstash	Real-time data caching
AI Agents	Claude API (tool use)	Macro, Liquidity, Sentiment, Cycle, Daily Brief agents
Frontend	React + Recharts	Dashboard, charts, chat interface, alerts
Hosting MVP	Railway + Neon.tech	~\$15-20/month (personal use)
Hosting SaaS	Railway + Supabase Pro	~\$100-130/month — break-even at 6 users x \$20/mo

All FREE sources are public APIs — no key required or free key available. FRED API key is free: register at fred.stlouisfed.org. SEC EDGAR API is fully public and requires no authentication.